The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 August 1998.

INCH-POUND

MIL-PRF-19500/461C
29 May 1998
SUPERSEDING
MIL-S-19500/461B
3 June 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, HIGH-POWER
TYPE 2N6211, 2N6212, 2N6213, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon, high-voltage. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (TO-66) and figure 2 JANHC and JANKC (die) dimensions.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{OP} and T_{STG}	$R_{\theta JC}$ (max)
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N6211	3.0	35	275	225	6.0	1.0	2.0	-55 to +200	5.0
2N6212	3.0	35	350	300	6.0	1.0	2.0	-55 to +200	5.0
2N6213	3.0	35	400	350	6.0	1.0	2.0	-55 to +200	5.0

1/ Derate linearly at 17.1 mW/° C for T_A > +25° C.
2/ Derate linearly at 200 mW/° C for T_C > +25° C.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25° C.

	h_{FE1} 1/	$V_{CE(SAT)}$ 1/			C_{obo}	$\|h_{fe}\|$	Pulse response	
	V_{CE} = 5 V dc	I_C = 1.0 A dc, I_B = -0.125 A dc			100 kHz ≤ f ≤ 1 MHz, V_{CB} = 10 V dc	f = 5 MHz, I_C = 0.2 A dc	t_{on}	t_{off}
	I_C = 1 A dc	2N6211	2N6212	2N6213	I_E = 0	V_{CE} = 10 V dc		
		V dc	V dc	V dc	pF		µs	µs
Minimum	30					4		
Maximum	175	1.4	1.6	2.0	220	20	0.6	3.1

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (T0-66) and 2 (die) herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein.

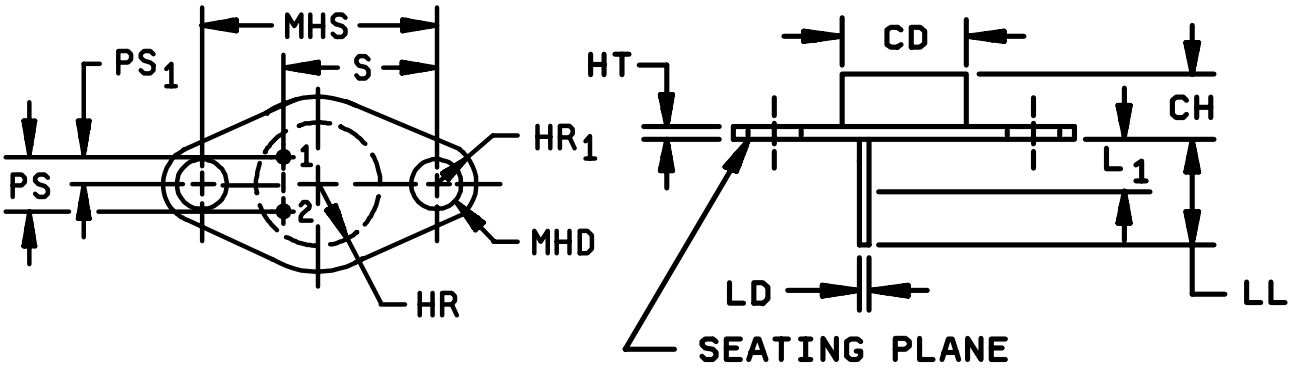
3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

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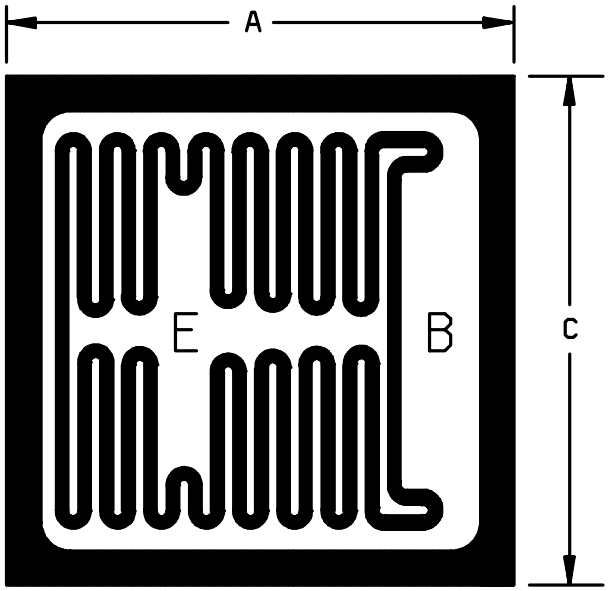


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.250	.340	6.35	8.64	
LD	.028	.034	0.71	0.86	7,9
CD	.470	.500	11.94	12.70	2
PS	.190	.210	4.83	5.33	3
PS_1	.093	.107	2.36	2.72	3
HT	.050	.075	1.27	1.91	2, 5
LL	.360	.500	9.14	12.70	7
L_1		.050		1.27	4
MHD	.142	.152	3.61	3.86	
MHS	.958	.962	24.33	24.43	
HR		.350		8.89	
HR_1	.115	.145	2.92	3.68	
S	.570	.590	14.48	14.99	3

NOTES:
1. Metric equivalents are given for general information only.
2. Body contour is optional within zone defined by CD.
3. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used, measurement shall be made at seating plane.
4. Within this zone the lead diameter may vary to allow for lead finishes and irregularities.
5. HT dimension does not include sealing flanges.
6. The seating plane of header shall be flat within .001 inch (0.025 mm), concave to .004 inch (0.101 mm), convex inside a .520 inch (13.20 mm) diameter circle on the center of the header, and flat within .001 inch (0.025 mm), concave to .006 inch (0.152 mm), convex overall.
7. Both terminals.
8. The collector shall be electrically connected to the case.
9. LD applies between L_1 and LL. Diameter is uncontrolled in L_1.
10. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.119	.125	3.02	3.18	
B	.119	.125	3.02	3.18	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Thickness: .006 inch (0.15 mm) to .012 inch (0.30 mm).
   Top metal: Aluminum 50,000 Å nominal, 37,500 Å minimum.
   Back metal: Gold 3,000 Å nominal.
   Back side: Collector.
   Bonding pad: B = .015 inch (0.38 mm) x .072 inch (1.83 mm).
   E = .015 inch (0.38 mm) x .060 inch (1.52 mm).
4. Junctions passivated with thermal silicon dioxide.

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	h_{FE1} and I_{CEX}	
11	h_{FE1} and I_{CEX}; ΔI_{CEX} = ±100 percent of initial value or 5 μA dc, whichever is greater. Δh_{FE1} = ±15 percent.	h_{FE1} and I_{CEX}
12	Burn-in (see 4.3.1)	Burn-in (see 4.3.1)
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX} = 100 percent of initial value or 5 μA dc, whichever is greater. Δh_{FE1} = ±15 percent	Subgroup 2 of table I herein; ΔI_{CEX} = 100 percent of initial value or 0.1 mA dc, whichever is greater. Δh_{FE1} = ±25 percent.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: MIL-STD-750, method 1038, test condition B.

T_A = room ambient as defined in the general requirements in 4.5 of MIL-STD-750; $V_{CB} \geq$ 20 V dc, P_T = 3.0°W.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 JANHC and JANKC screening. Screening of die shall be in accordance with MIL-PRF-19500. Test limits and conditions shall be chosen by the supplier to demonstrate compliance with electrical characteristics specified by detail specification. Probe test shall be performed 100 percent by the supplier on the entire die lot.

4.3.3 JANHC and JANKC die. Qualification shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	$V_{CB} \geq$ 10 V dc, ΔT_J between cycles ≥ +100°C, $t_{on} = t_{off} \geq$ 3 minutes for 2,000 cycles. Forced air cooling on the devices shall be permitted only during the t_{off} time.
B5	1027	$V_{CE} \geq$ 10 V dc, $T_A \leq$ +125°C, adjust T_A and power to achieve a $T_J \geq$ +275°C, t = 96 hours.

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4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1037	$V_{CB} \geq 10$ V dc, ΔT_J between cycles $\geq +100°C$, $t_{on} = t_{off} \geq 3$ minutes for 2,000 cycles. Forced air cooling on the devices shall be permitted only during the t_{off} time.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 10 pounds, t = 15 seconds.
C6	1037	$V_{CB} \geq 10$ V dc, ΔT_J between cycles $\geq +100°C$, $t_{on} = t_{off} \geq 3$ minutes for 6,000 cycles. Forced air cooling on the devices shall be permitted only during the t_{off} time.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 1.0 A dc.

b. Collector to emitter voltage magnitude shall be $\geq$ 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit shall be $R_{\theta JC} = 5.0°C/W$.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 200 mA dc, pulsed (see 4.5.1), (or L = 10 mH, f = 30-60 Hz)	$V_{(BR)CEO}$			
2N6211				225		V dc
2N6212				300		V dc
2N6213				350		V dc
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 200 mA dc, pulsed (see 4.5.1), (or L = 10 mH, f = 30-60 Hz) R_{BE} = 50 Ω	$V_{(BR)CER}$			
2N6211				250		V dc
2N6212				325		V dc
2N6213				375		V dc
Collector to emitter breakdown voltage	3011	Bias condition D, I_C = 200 mA dc, pulsed (see 4.5.1), (or L = 10 mH, f = 30-60 Hz) V_{BE} = 1.5 V dc, R_{BE} = 50 Ω	$V_{(BR)CEX}$			
2N6211				275		V dc
2N6212				350		V dc
2N6213				400		V dc
Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 150 V dc	I_{CEO}		5.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX1}		0.5	mA dc
2N6211		V_{CE} = 250 V dc				
2N6212		V_{CE} = 315 V dc				
2N6213		V_{CE} = 360 V dc				
Emitter to base voltage cutoff current	3061	Bias condition D, V_{EB} = 6 V dc	I_{EBO}		0.5	mA dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO}		15	mA dc
2N6211		V_{CB} = 275 V dc				
2N6212		V_{CB} = 350 V dc				
2N6213		V_{CB} = 400 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE1}	10	100	
2N6211		V_{CE} = 2.8 V dc				
2N6212		V_{CE} = 3.2 V dc				
2N6213		V_{CE} = 4.0 V dc				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N6211				30	175	
2N6212				30	175	
2N6213				30	150	
Collector to emitter saturated voltage	3071	I_C = 1 A dc, I_B = 0.125 A dc, pulsed (see 4.5.1)	$V_{CE(sat)}$			
2N6211					1.4	V dc
2N6212					1.6	V dc
2N6213					2.0	V dc
Base to emitter saturated voltage	3066	Test condition A, I_C = 1 A dc, I_B = 0.125 A dc, pulsed (see 4.5.1)	$V_{BE(sat)}$		1.4	V dc
Subgroup 3						
High temperature operation:		T_A = +100°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX2}		5	mA dc
2N6211		V_{CE} = 250 V dc				
2N6212		V_{CE} = 315 V dc				
2N6213		V_{CE} = 360 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc, I_C = 1 A dc, pulsed (see 4.5.1)	h_{FE3}	10		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Pulse response	3251	Test condition A, except test circuit and pulse requirements in accordance with figure 3 herein.				
Turn-on time		V_{CC} = 200 V dc ±10 V dc, I_C = 1 A dc, I_{B1} = -0.125 A dc	t_{on}		0.6	µs
Turn-off time		V_{CC} = 200 V dc ±10 V dc, I_C = 1 A dc, I_{B1} = -0.125 A dc, I_{B2} = 0.125 A dc	t_{off}		3.1	µs
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 0.2 A dc, f = 5 MHz	$\lvert h_{fe} \rvert$	4	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}		220	pF
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C, t = 1 s, 1 cycle (see figure 4)				
Test 1 (all device types)		I_C = 2.0 A dc, V_{CE} = 17.5 V dc				
Test 2 (all device types)		I_C = 0.875 A dc, V_{CE} = 40 V dc				
Test 3						
2N6211 only		I_C = 0.034 A dc, V_{CE} = 225 V dc				
Test 4						
2N6212 only		I_C = 0.02 A dc, V_{CE} = 300 V dc				
Test 5						
2N6213 only		I_C = 0.015 A dc, V_{CE} = 350 V dc				
Electrical measurements		See table I, group A, subgroup 2				

See footnote at end of table.

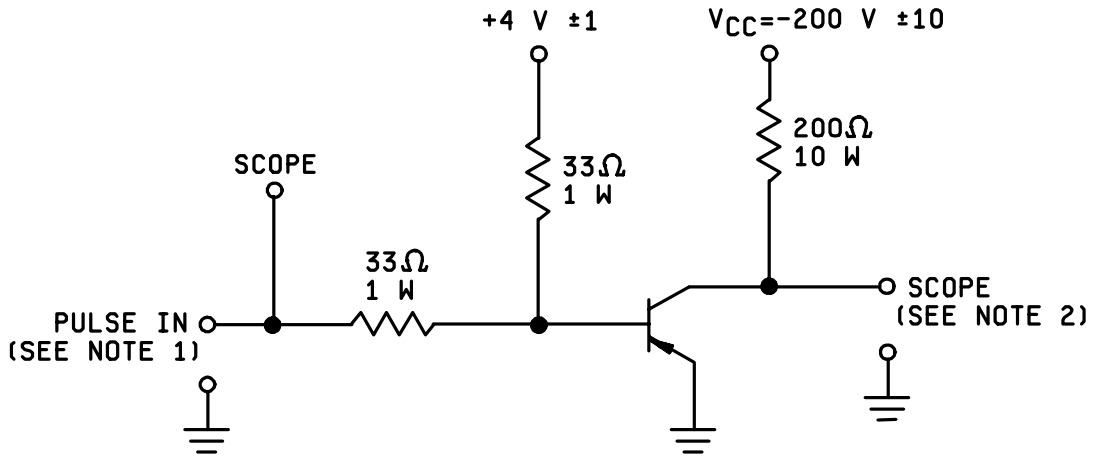
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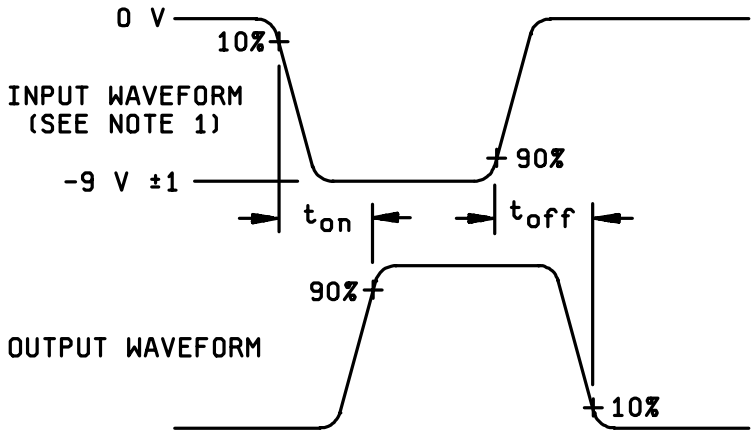
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Safe operating area (switching)	3053	Load condition C (unclamped inductive load) (see figure 5) T_C = +25°C, duty cycle ≤ 10 percent				
Test 1		t_p ≈ 15 ms, R_{BB1} = 20 Ω, V_{BB1} = 10 V dc maximum, R_{BB2} = 20 Ω, V_{BB2} = 1.5 V dc, V_{CC} = 10 V dc, I_C = 2 A dc, L = 25 µH, R_S = 0.1 Ω				
Test 2		t_p ≈ 15 ms, R_{BB1} = 50Ω, V_{BB1} = 10 V dc maximum, R_{BB2} = 20 Ω, V_{BB2} = 1.5 V, V_{CC} = 10 V dc, I_C = 0.25 A dc, L = 3.0 mH, R_S = 1.0 Ω				
Electrical measurements		See table I, group A, subgroup 2				
Safe operating area (switching)	3053	Load condition B, (clamped inductive load) see figure 6, T_C = +25°C, duty cycle ≤ 10 percent, t_p ≈ 15 ms, R_S = 0.1Ω, R_{BB1} = 20 Ω, R_{BB2} = 20 Ω, V_{BB1} = 10 V dc maximum, V_{BB2} = 1.5 V, I_C = 2.0 A dc (clamped voltage must be reached)				
2N6211		V_{CC} = 225 V dc, R_L ≤ 112.5Ω				
2N6212		V_{CC} = 300 V dc, R_L ≤ 150 Ω				
2N6213		V_{CC} = 350 V dc, R_L ≤ 175 Ω, L = 250 µH, CR = 1N1190A				
2N6211 2N6212 2N6213		Clamp voltage = 225, +0, -5 V dc, Clamp voltage = 300, +0, -5 V dc Clamp voltage = 350, +0, -5 V dc				
Electrical measurements		See table I, group A, subgroup 2				

1/ For sampling plan, see MIL-PRF-19500.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 20 ns, duty cycle ≤ 2 percent, generator source impedance shall be 50 Ω; pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 50 pF, rise time ≤ 20 ns.

FIGURE 3. Pulse response test circuit.

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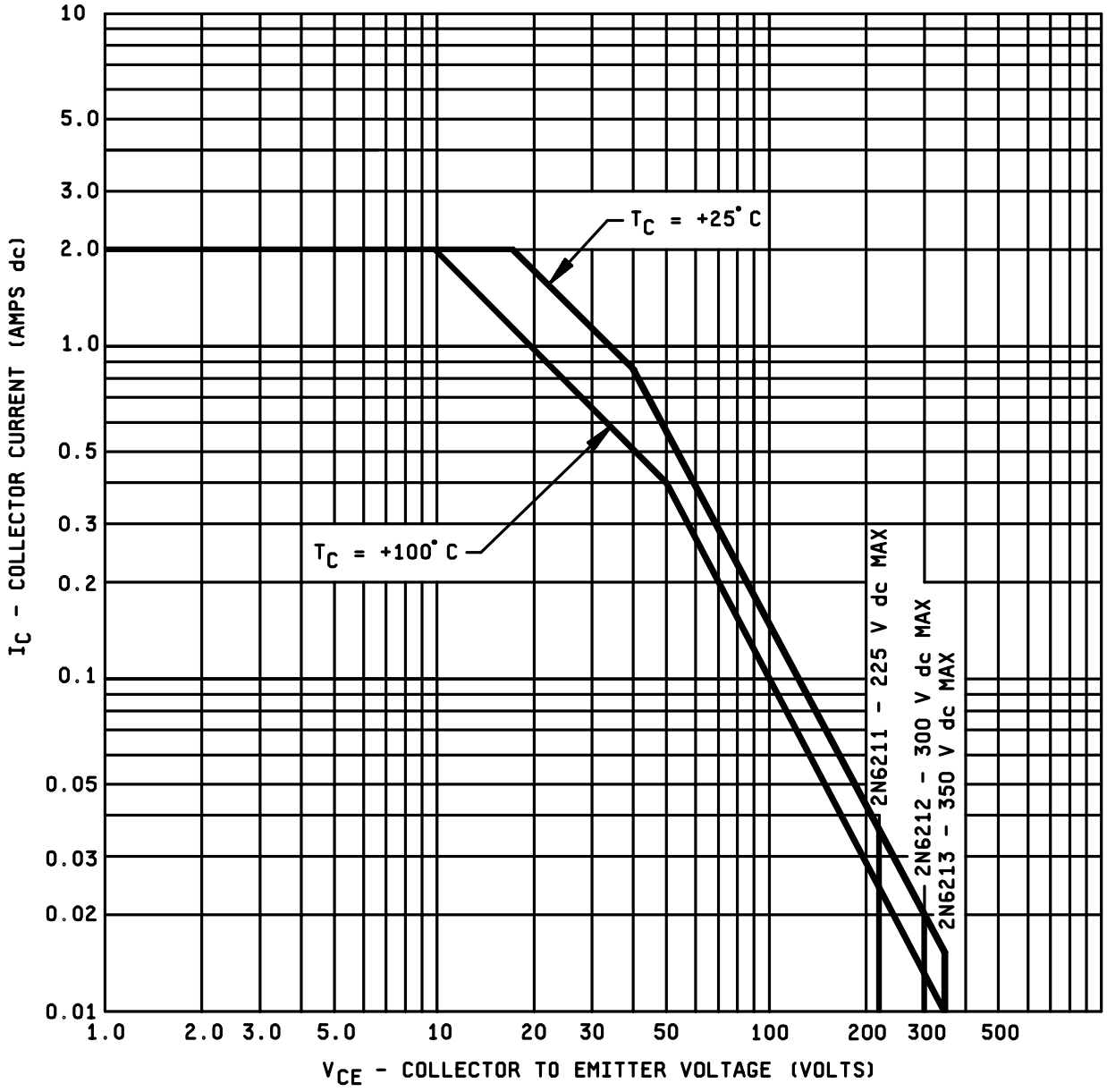


FIGURE 4. Maximum safe operating graph (continuous dc).

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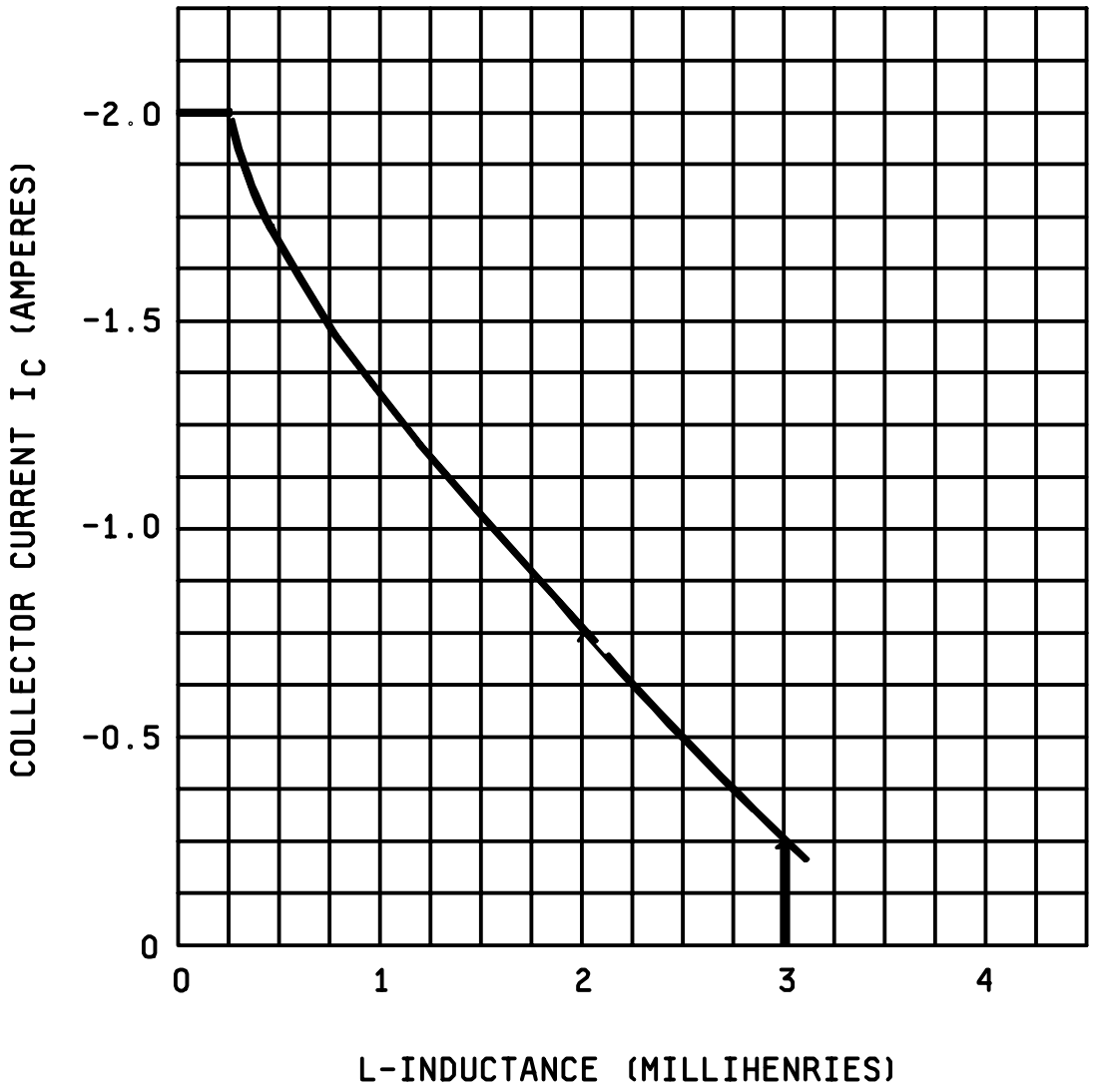


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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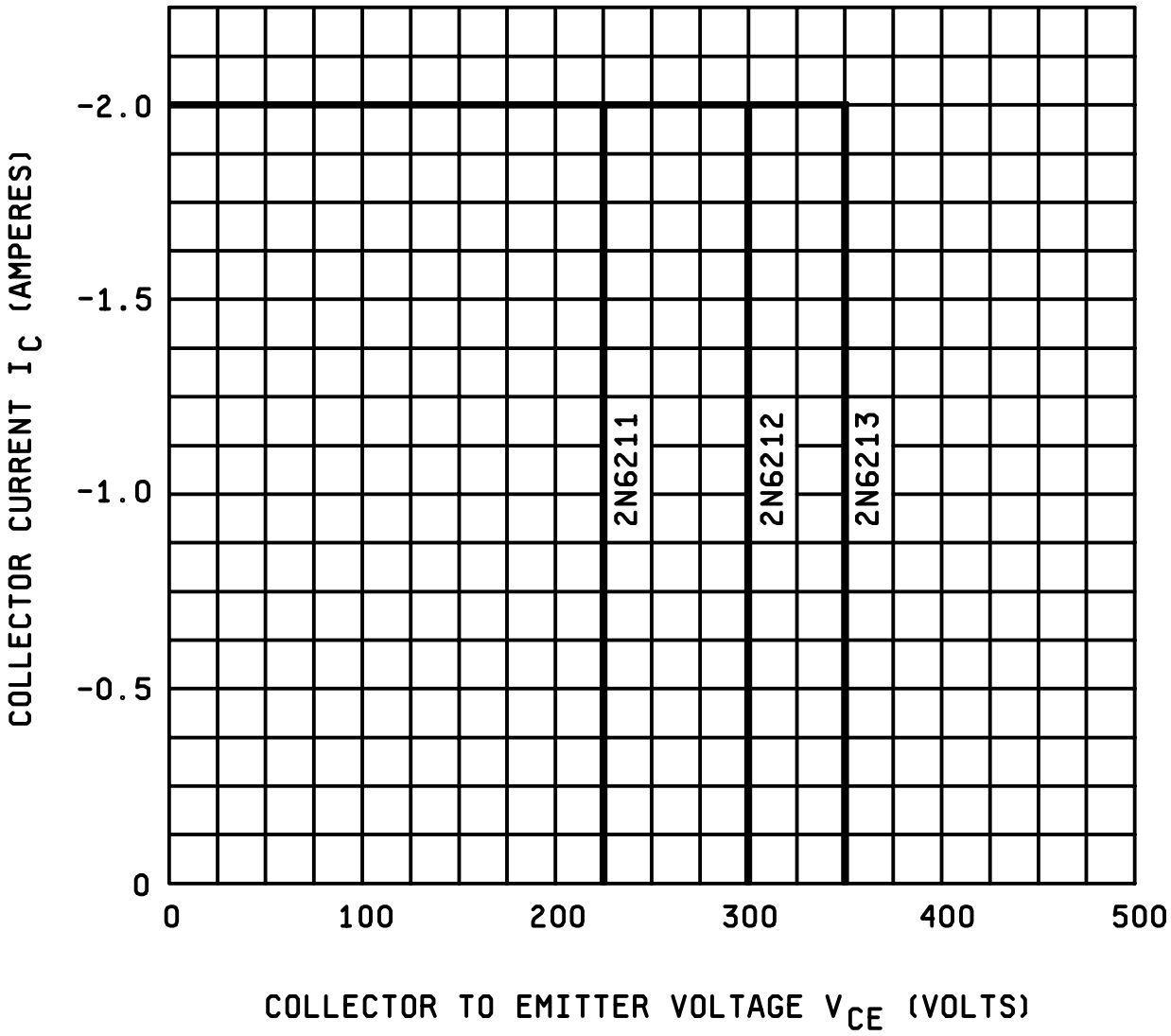


FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Type designation and quality product assurance level.

c. Product assurance and type designation, and for die acquisition the JANHC and JANKC identification (see figure 2 and 6.3).

6.3 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N6211) will be identified on the QPL.

JANC ordering information	
PIN	Manufacturer
	33178
2N6211	JANHCA2N6211 JANKCA2N6211
2N6212	JANHCA2N6212 JANKCA2N6212
2N6213	JANHCA2N6213 JANKCA2N6213

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Reviewing activities:
Army - AV
Air Force - 19, 85, 99

Preparing activity:
DLA- CC

(Project 5961-1810)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/461C	2. DOCUMENT DATE 29 May 1998

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, HIGH-POWER TYPE 2N6211, 2N6212, 2N6213, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete

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